ZEESHAN MEMON

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EDUCATION

Ph.D. Computer Science, Emory University 2024 - Ongoing
Supervised By: Dr. Liang Zhao
Research Interests: Foundation Models for Complex Networks, Optimization, Multi-Agent Communication

Bachelors of Software Engineering, National University of Science and Technology 2020- 2024
CGPA: 3.99/4.0 (Class Rank: 1/90)
Research Specialization: Deep Learning, Computer Vision & Representation Learning
Research Assistant advised by Dr. Faisal Shafait

EXPERIENCE

Research Intern May 2026 – Aug 2026
Microsoft Research *Redmond, WA, USA*

- Working on agentic systems for infrastructure planning tasks, i.e. engineering workflows.
- Specifically building memory mechanisms to help agents self-improve under compute constraints.

Visiting Student Oct 2025 – Present
Argonne National Laboratory *Remote*

- Working on foundation models for large-scale power-grid optimization(ACOPF, SCUC, DSSE and other related problems), integrating constraint-aware, learning-augmented solvers as part of the Genesis Mission Project.
- Developing scalable training and evaluation pipelines for ML-based optimization models, with systematic benchmarking of homogeneous and heterogeneous models.
- Conducting large-scale experiments on HPC environments, combining graph-based ML models with mathematical optimization solvers.
- Designing workflows for integration of learned models within agentic framework(part of GridMind Project at ANL).
- Advised by: Dr. Kibaek Kim

OpenLab Summer Student Jun 2023 – Aug 2023
CERN *Geneva, Switzerland*

- Developed generative models for highly sparse simulation data in high-energy physics(Geant IV simulation data).
- Investigated loss modeling and training strategies for transformer-based generative models to improve stability and sample quality.
- Supervised by: Dr. Dalila Salamani and Piyush Raikawar

Research Fellow Jun 2022 – Sep 2022
Hoschule RheinMain *Wiesbaden, Hessen, Germany*

- Benchmarked deep generative models for satellite image super-resolution across multiple datasets and architectural choices.
- Explored multi-modal spatial fusion techniques to improve reconstruction quality in super-resolution tasks.
- Supervised by: Dr. Adrian Ulges and Dr. Ulrich Schwanecke

PUBLICATIONS

- **Towards Systematic Generalization for Power Grid Optimization Problems** *KDD, 2026 (Oral)*
  Zeeshan Memon, *Yijiang Li, Hongwei Jin, Kibaek Kim, Liang Zhao*
  ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), **Oral Presentation**, arXiv:2605.02026

- **EpiWorld: Grounding LLM Policy Agents in Epidemiological World Models** *EMNLP Findings, 2026*
  ***Zeeshan Memon**, Yiqi Su, Kai Shu, Naren Ramakrishnan, Liang Zhao*
  Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP Findings)
- **CONSTRAINER: Promptable Graph-Structured Optimization via Constraint Conditioning** *Under Review*
  ***Zeeshan Memon**, Mingke Tian, Xinyuan Song, Hongwei Jin, Kibaek Kim, Liang Zhao*
  Currently under review
- **Deep Identification of Propagation Trees** *IJCAI, 2026*
  ***Zeeshan Memon**, Chen Ling, Ruochen Kong, Vishwanath Seshagiri, Andreas Zufle, Liang Zhao*
  International Joint Conference on Artificial Intelligence (IJCAI), arXiv:2503.00646
- **LUMINA: Foundation Models for Topology Transferable ACOPF** *ICLR Workshop, 2026*
  *Yijiang Li, **Zeeshan Memon**, Hongwei Jin, Stefano Fenu, Keunju Song, Sunash B. Sharma, Parfait Gasana, Hongseok Kim, Liang Zhao, Kibaek Kim*
  ICLR 2026 Workshop on Foundation Models for Science (FM4Science), arXiv:2603.04300
- **Toward World Models for Epidemiology** *ICLR Workshop, 2026*
  ***Zeeshan Memon**, Yiqi Su, Christo Kurisummoottil Thomas, Walid Saad, Liang Zhao, Naren Ramakrishnan*
  ICLR 2026 Workshop on World Models, arXiv:2604.09519
- **On the Fundamental Limits of LLMs at Scale** *TMLR, 2025*
  *Muhammad Ahmed Mohsin, Muhammad Umer, Ahsan Bilal, **Zeeshan Memon**, . . . , Muhammad Ali Jamshed, John M. Cioffi*
  Transactions on Machine Learning Research (TMLR), arXiv:2511.12869
- **Deep Causal Generative Models with Property Control** *Preprint, 2025*
  *Qilong Zhao, Shiyu Wang, **Zeeshan Memon**, Yang Qiao, Guangji Bai, B Pan, Zhaohui Qin, Liang Zhao*
  arXiv:2405.16219v2
- **LLM-Informed Discrete Prompt Optimization** *ICML Workshop, 2024*
  ***Zeeshan Memon**, Muhammad Arham, Adnan Ul-Hasan, Faisal Shafait*
  ICML 2024 Workshop on LLMs and Cognition
- **Content-Aware Urdu Handwriting Generation** *ICDAR, 2023*
  ***Zeeshan Memon**, Adnan Ul-Hasan, Faisal Shafait*
  Proceedings of the 17th International Conference on Document Analysis and Recognition (ICDAR), San José, CA, USA

ACHIEVEMENTS

- Received KDD 2026 Student Travel Award
- Awarded Rector's Gold Medal for best final-year undergraduate research project
- Awarded Presidential Gold Medal for securing rank one in batch in Bachelors of Software Engineering
- Selected as prestigious Openlab Researcher at CERN from pool of 2000 candidates (1.5% acceptance rate).
- Awarded prestigious MBZUAI undegraduate fully funded research internship.
- Awarded DAAD fully funded research internship.
- Got NUST Merit Based Scholarship for 8 consecutive semesters

SERVICES

Reviewing & Program Committee

Reviewer, NeurIPS 2026 · Reviewer, KDD 2026 · PC Member, LLMs4Science @ AAAI 2025 · PC Member, IEEE International Conference on Data Science and Advanced Analytics (DSAA) 2025 · Reviewer, IEEE Transactions on Big Data · Reviewer, IEEE Transactions on Computational Social Systems

Teaching Experience

Teaching Assistant for 584: Spatial Computing@ Emory University
Teaching Assistant for CS584: Deep Learning on Graphs@ Emory University
Teaching Assistant for CS 110: Computer Science Fundamentals @ Emory University
Teaching Assistant for CS 250: Data Structures and Algorithms @ SEECS, NUST